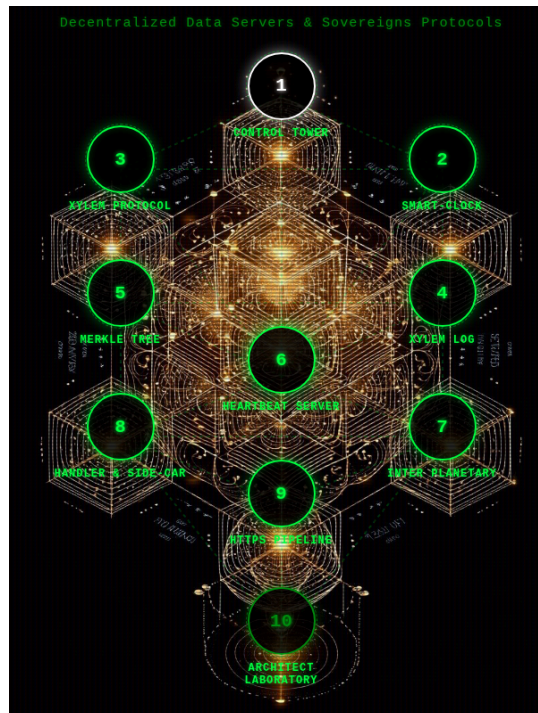




The Big Tree Universal Harmonic System

Architected by Oliver Big Tree — Artist & Engineer

The **Big Tree Universal Harmonic System** is a dynamic, isomorphic digital organism designed to generate, notarize, and permanently store an immutable daily Oracle.

While the underlying architecture combines Bitcoin, IPFS, Ethereum Layer-2, and Nostr, the core mechanic is simple: **24 hourly snapshots, fully proven, cryptographically sealed every 24 hours, and accessible through a single event.**

How the Daily Oracle Workflow Operates

[Module 1: Image Stream]

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▼ (Every Hour)

[Module 2: IPFS Indexing (CID) + Bitcoin OpenTimestamps (OTS)]

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▼ (Throughout the Day)

[Local Handler: Transaction Recording]

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▼ (End of Day)

[Compiler: Chain Concatenation + L2 Ethereum Smart Contract]

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▼ (Final Step)

[GitHub Action: Encrypted Nostr Event Release (24 OTS + Images)]

The 3-Phase Lifecycle

1. Hourly Capture & Bitcoin Notarization

- **Continuous Generation:** Module 1 generates visual feeds continuously.
- **Hourly Capture:** Every hour, Module 2 extracts a single image frame and generates its IPFS content identifier (CID).
- **Time-Stamping:** This image CID is immediately anchored onto the Bitcoin blockchain via OpenTimestamps (OTS), creating an unfalsifiable proof of existence.

2. Daily Aggregation & Layer-2 Sealing

- **Local Ledger:** Throughout the day, all system transactions log into a local handler.
- **Custom Blockchain:** At day's end, a custom compiler merges the daily transactions with the Oracle data, concatenating everything into an immutable ledger.
- **Ethereum L2 Settlement:** Automated processes submit a smart contract transaction on an Ethereum L2 network containing the 24 hourly CIDs, the current Day Hash (\$H_{day}\$), and the Previous Parent Hash (\$H_{parent}\$)—permanently sealing the workflow.

3. The Nostr Payload (The Core Innovation)

Instead of cluttering the network with multiple entries, the final step unifies the entire day into one lightweight reference point:

- **Encrypted Relay Broadcast:** A automated GitHub Action retrieves the 24 daily OTS files from our Swiss server, encrypts them, and broadcasts them as a single payload across the Nostr network.
- **Single Event ID Access:** Through one compact Nostr **Event ID**, anyone can query the relays, retrieve all 24 OTS cryptographic proofs, and reconstruct/verify the 24 visual frames.
- **Web3 Gateway (Day +1):** A dedicated Web3 gate publishes a daily HTML page displaying the Day Hash (\$H_{day}\$) alongside interactive console tools for full public verification.

Key System Highlights

- **Single Event ID Portability:** Access 24 high-resolution image proofs through one concise Nostr hash.
- **Triple-Layer Trust:** Powered by Bitcoin (PoW Time-Stamping), Ethereum L2 (Smart Contract Execution), and IPFS/Nostr (Decentralized Storage & Messaging).
- **Full Transparency:** Anyone with a standard web console can inspect the daily hashes, verify the OTS files, and confirm the Oracle's integrity.

Hot Links : https://bigtreeconnection.com/web3/daily_log.html